SIDARTH SAI

+91-938-435-6156 | sidarthsai13901@gmail.com | Chennai, Tamil Nadu, India

OVERVIEW

B.Tech Electronics and Communication Engineering graduate with approximately 3 years of combined experience in embedded systems across firmware development, hardware/software integration, vehicle control systems, and Linux-based embedded applications. Hands-on experience with Embedded C, MSP432, STM32, peripheral interfaces, clock configuration, debugging tools, and Python/Linux utilities. Also experienced in coordinating with external vendors, clarifying technical specifications, and following up on engineering issues. Interested in applying this foundation to technical support and embedded semiconductor applications.

EDUCATION

B.Tech, Electronics and Communication Engineering — Amrita Vishwa Vidyapeetham | Jul 2019 – Jul 2023 | Chennai, India

CBSE Class 12 — Velammal Vidhyashram | Mar 2019 | Chennai, India

CBSE Class 10 — Velammal Vidhyashram | Mar 2017 | Chennai, India

PROFESSIONAL DEVELOPMENT

Post Graduate Diploma in Electric Vehicles and Embedded Systems — DIY Guru | Ongoing

AUTOSAR Bootcamp — Global Scalar | Ongoing

EXPERIENCE

Snerith Engineers — Embedded and Control System Engineer

Chennai, India | Oct 2025 – Present

- Analyzed vehicle software requirements across sensing, control, telematics, HMI, safety, and emergency functions, allocating functional responsibilities to embedded controllers and devices.
- Contributed to vehicle E/E and software architecture by defining signal flows and interfaces among HMI, VCU/controllers, CAN networks, sensors, and actuators; derived control requirements from vehicle performance and motor-sizing calculations.
- Developed and customized an Embedded Linux platform on the Freescale/NXP i.MX6 ARM SoC, including platform configuration, Device Tree/peripheral configuration, and integration of hardware interfaces where applicable to the project.
- Performed hardware/software integration activities, validating UART, SPI, I²C, GPIO, Ethernet, USB, and CAN; used logs and system interfaces to investigate initialization and runtime issues.
- Developed Qt/QML-based HMI applications and supported integration between application interfaces, vehicle functions, and embedded hardware.
- Designed a low-cost physical test bench for vehicle-control software validation using development boards and test equipment to emulate vehicle inputs, HMI commands, hydraulic behavior, auxiliary loads, and CAN interfaces; implemented GPIO/relay control for lighting, horn, and warning buzzer.
- Developed model-based differential steering control using potentiometer feedback, closed-loop hydraulic clamp control using pressure feedback, and synchronized hydraulic lifting/lowering using position feedback.
- Coordinated with external vendors on electrical/electronic components and engineering deliverables, clarified technical specifications and interfaces, tracked progress, and followed up on technical queries with internal stakeholders.

6G Testbed, IIT Madras — Project Associate, Embedded Systems

Chennai, India | Jan 2024 – Aug 2025

- Developed Embedded C firmware on MSP432 using TI DriverLib to interface with INA237 and INA219 power-monitoring ICs and TDK FS1525 devices, collecting measurements and streaming data over UART.
- Worked with SPI/I²C device communication and peripheral configuration, using device documentation and interface behavior to support integration and debugging.
- Developed a Qt/QML-based Raspberry Pi/Linux utility for configuration of timing hardware used in 5G/6G testbed applications, including Ethernet-based transfer of hexadecimal configuration files and programming of Si5518/Si5348 timing devices through SPI/I²C.
- Implemented a remote configuration workflow to apply updated timing-device configuration files over Ethernet without modifying the application software.
- Developed Python/Linux utilities on an FPGA-SoC platform for UART-based u-blox GPS/GNSS communication, including NMEA message parsing and validation.
- Integrated and debugged Ethernet, UART, SPI, I²C, GPIO, and USB interfaces across ARM-based embedded Linux and FPGA-SoC platforms; used serial logs, board documentation, datasheets, and test equipment to investigate peripheral and system-integration issues.

TECHNICAL SKILLS

Embedded Programming: Embedded C, C++, Python, shell scripting, bare-metal firmware, peripheral drivers, interrupts, DMA.

Microcontrollers & Processors: MSP432, STM32F4 (ARM Cortex-M4), TI Hercules (ARM Cortex-R4F), Freescale/NXP i.MX6, Raspberry Pi 5, Infineon AURIX TC334, FPGA-SoC platforms.

Peripherals & Communication: GPIO, ADC, PWM, timers, UART, SPI, I²C, CAN, CAN-FD, Ethernet, USB, clocking and timing devices.

Embedded Linux: Linux-based application development, Raspberry Pi 5 BSP and hardware bring-up, Linux kernel/device-tree configuration, boot flow familiarity, Python/Linux utilities, and peripheral integration.

Operating Systems: FreeRTOS, Embedded Linux; familiarity with QNX.

IDE & Toolchains: STM32CubeIDE, Code Composer Studio, Keil, ARM GCC, GCC, TI DriverLib, AURIX Development Studio, Git.

Debugging & Test: GDB, SWD/JTAG, ST-LINK, XDS110, miniWiggler/DAP, oscilloscope, logic analyzer, serial debugging, Waveshare USB-CAN-FD-B.

Application & Diagnostics Tools: Qt/QML, python-can, cantools, can-isotp, udsoncan, Kvaser CanKing, Kvaser Database Editor.

Additional Knowledge: AUTOSAR Classic, UDS diagnostics, MATLAB, Simulink, Stateflow, Simscape, vehicle E/E architecture, model-based development.

SELECTED PROJECTS

STM32F446RE Bare-Metal BSP & Hardware Bring-Up

- Developed a register-level BSP including startup code, linker configuration, clock initialization, GPIO, and peripheral abstraction; used datasheets, reference manuals, and schematics to understand memory mapping, clock trees, pin configuration, and interfaces.
- Implemented and validated UART, SPI, I²C, ADC, PWM, timers, CAN, and USB using interrupt/DMA-based operation and SWD debugging.

Raspberry Pi 5 Linux BSP & Hardware Bring-Up

- Configured and investigated the Raspberry Pi 5 Linux boot flow, including bootloader, kernel, Device Tree, system image, and hardware initialization.
- Reviewed board interfaces and configured GPIO, UART, I²C, SPI, memory, and peripheral settings through Linux device configuration.
- Troubleshot boot, kernel, driver, and peripheral issues using the serial console, kernel logs, GDB, and hardware-level diagnostics.

AURIX TC334 AUTOSAR Classic Communication & Diagnostics ECU — Ongoing

- Working on CAN-FD communication and diagnostic flows using AUTOSAR Classic concepts, including communication, diagnostic services, DTC handling, timeout checks, and simulated fault-recovery validation.

AURIX TC334 FreeRTOS Multicore Processing System — Ongoing

- Developing a multicore application with inter-core communication, shared-memory exchange, synchronization, and analysis of task scheduling and communication latency.